WORKED SOLUTION

STEP I 2002, Question 2

Pure • Curve Sketching

STEP I 2002, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q2

The key idea

The polynomial $x^m(x-1)^n$ vanishes at $x=0$ and $x=1$, so something must turn round in between. Differentiate with the product rule and *take out every common factor*: the derivative then splits into simple pieces whose signs we can read off. The whole sketch is controlled by parity: whether a power is even or odd decides whether the curve touches the axis or crosses it, and which side of the axis it is on.

Solution

The stationary point

By the product rule,

$$\begin{aligned} \mathrm{f}'(x) &= mx^{m-1}(x-1)^n + nx^m(x-1)^{n-1} \\ &= x^{m-1}(x-1)^{n-1}[m(x-1)+nx] \\ &= x^{m-1}(x-1)^{n-1}[(m+n)x-m]. \end{aligned}$$

So $\mathrm{f}'(x)=0$ exactly when $x=0$, $x=1$ or

$$x = x_0 = \frac{m}{m+n}.$$

Since $m, n > 0$ we have $0 < m < m+n$, so $0 < x_0 < 1$. That is the required stationary point. (Since $m, n \geq 2$, the powers $m-1$ and $n-1$ are at least 1, so $x=0$ and $x=1$ are stationary points too. We will need them for the sketches.)

Its nature

Write $\mathrm{f}'(x) = g(x)\,h(x)$ with $g(x) = x^{m-1}(x-1)^{n-1}$ and $h(x) = (m+n)x - m$. Then

$$\mathrm{f}''(x) = g'(x)h(x) + g(x)h'(x) = g'(x)h(x) + (m+n)g(x).$$

Ask yourself Do we really need to differentiate g in full?

No. At x_0 we have $h(x_0) = 0$, so the first term vanishes, whatever g' is:

$$\mathrm{f}''(x_0) = (m+n)\, x_0^{m-1}\,(x_0-1)^{n-1}.$$

Here $m+n > 0$ and $x_0^{m-1} > 0$, but $x_0 - 1 < 0$. So $\mathrm{f}''(x_0)$ is nonzero, with the sign of $(-1)^{n-1}$:

- if n is even, $n-1$ is odd, $\mathrm{f}''(x_0) < 0$ and x_0 is a **maximum**;
- if n is odd, $n-1$ is even, $\mathrm{f}''(x_0) > 0$ and x_0 is a **minimum**.

Because $\mathrm{f}''(x_0) \neq 0$, the second-derivative test is conclusive. That is what we had to show.

The sketches

The four cases are given by the parities of m and n. To sketch each one, we collect these facts.

Sign of f. For $x > 1$ both factors are positive, so $\mathrm{f} > 0$. For $0 < x < 1$, $x^m > 0$ and $(x-1)^n$ has the sign of $(-1)^n$. For $x < 0$, the sign is $(-1)^m(-1)^n = (-1)^{m+n}$.

Behaviour at the roots. Near $x = 0$ we have $\mathrm{f}(x) \approx (-1)^n x^m$. If m is even, f keeps its sign either side of 0, so the curve touches the axis there (a turning point). If m is odd, f changes sign, and because $\mathrm{f}'(0) = 0$ the curve crosses the axis horizontally (a stationary point of inflection). The same reasoning with n applies at $x = 1$.

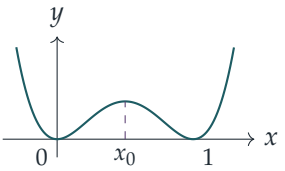
Ends. $\mathrm{f}(x)$ behaves like x^{m+n} for large $|x|$, so $\mathrm{f} \to +\infty$ as $x \to +\infty$. As $x \to -\infty$, $\mathrm{f} \to +\infty$ if $m+n$ is even and $\mathrm{f} \to -\infty$ if $m+n$ is odd.

No other stationary points. The only ones are $x = 0$, x_0 and $x = 1$.

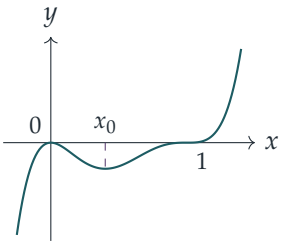
Putting these together:

- m **even,** n **even:** $\mathrm{f} \geq 0$ everywhere. There are minima on the axis at 0 and 1 and a maximum at x_0: a W shape.
- m **even,** n **odd:** the curve rises from $-\infty$ to touch the axis from below at 0 (a maximum), falls to the minimum at x_0, then rises through a stationary inflection at 1 and on to $+\infty$.
- m **odd,** n **even:** the curve rises from $-\infty$ through a stationary inflection at 0 to the maximum at x_0, falls to touch the axis at 1 (a minimum), then rises to $+\infty$.
- m **odd,** n **odd:** the curve comes down from $+\infty$, through a stationary inflection at 0 to the minimum at x_0, back up through a stationary inflection at 1, and on to $+\infty$.

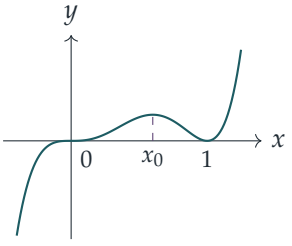
The sketches below use the smallest cases $(m,n) = (2,2), (2,3), (3,2), (3,3)$, each with its own vertical scale. The values at x_0 are small: $\frac{1}{16}$, $-\frac{108}{3125}$, $\frac{108}{3125}$ and $-\frac{1}{64}$.



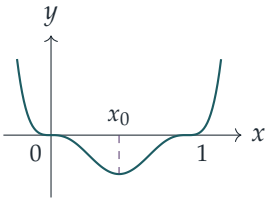
m even, n even



m even, n odd



m odd, n even



m odd, n odd

Checks

- Take $m = n = 2$: $\mathrm{f}(x) = \big(x(x-1)\big)^2$ is symmetric about $x = \frac{1}{2}$, and $x_0 = \frac{1}{2}$ is indeed a maximum, with value $\frac{1}{16}$.
- The substitution $x \mapsto 1 - x$ turns $x^m(x-1)^n$ into $(-1)^{m+n}x^n(x-1)^m$. So swapping m and n reflects the graph in $x = \frac{1}{2}$ (and also in the x-axis if $m + n$ is odd). This is why the two mixed-parity sketches are mirror images of each other.
- We checked f' against a numerical derivative, and the sign of $\mathrm{f}''(x_0)$ against a finite difference, for all $2 \leq m, n \leq 8$. For $(m, n) = (2, 3)$, for instance, $\mathrm{f}''(\frac{2}{5}) = 5 \cdot \frac{2}{5} \cdot \frac{9}{25} = 0.72 > 0$, a minimum, as predicted.

Common slips

- Expanding $\mathrm{f}''(x)$ in full. At x_0 the factor $(m+n)x - m$ is zero, so only one term survives.
- Forgetting that $x = 0$ and $x = 1$ are also stationary points (because $m, n \geq 2$), and drawing the curve crossing the axis at an angle there.
- Getting the sign for $x < 0$ wrong. It depends on $m + n$, not on m alone.

Beyond STEP

The local shape of a polynomial near a root of multiplicity k looks like $c(x-a)^k$, and the parity of k decides touching or crossing. This is the first step towards Taylor's theorem and the higher-derivative test for stationary points: if the first nonzero derivative at a stationary point has order k, the point is a turning point when k is even and an inflection when k is odd. Abbott's *Understanding Analysis* puts these ideas on a rigorous footing.

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